IHP SG13G2 · OPEN SOURCE RFIC EM WORKFLOW

EM Simulation from KLayout: User's Guide for the klayout_em_export Plug-in (Linux)

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- **Holes:** on every metal of the stackup (not the vias), an opening size found at least 20 times is cheesing and is filled, in any shape. Openings of other sizes belong to your circuit and stay.
- **Merge:** then the metal of the current cell is merged into as few polygons as possible. The metal of subcells placed at least twice (via stacks, MIM capacitors) is merged in too, and so is the metal of a subcell that holds filled holes (a cheesed plane in a cell of its own).
- **What stays:** devices, PDK cells, other subcells, cells that are also placed somewhere else, all vias (the export writes every via, and gds2palace corrects the via arrays itself), texts and the plug-in's own marked shapes.
- A message lists per metal the holes filled and the polygons before and after. On the 60 GHz PA of the examples: 2899 metal polygons before, 387 after.
- **Edit > Undo** takes it back. Once you save and close the layout, the change stays: **never tape out a converted layout**, the fill breaks the density rules. Convert a copy (File > Save As first) if you are unsure.

6 Ports

In this chapter: what an EM port is, and how to place one.

Why it matters: the S-parameters are only as good as the ports. A port defines where the EM simulation feeds signal into the layout and measures it, and between which two conductors. A port in the wrong place or with the wrong reference gives you a wrong reference plane, or a short circuit.

6.1 What is a port?

A port is a small source between two conductors: the signal conductor and its reference (ground). The plug-in uses two kinds:

- **Via port:** a vertical port between two metal layers, e.g. from the pad on TopMetal2 down to the ground plane on Metal5. You draw it as a line (zero width) along a metal edge. This is the usual port at a pad.
- **In-plane port:** a horizontal port across a gap between two shapes of the same metal, e.g. in a coplanar line. You draw it as a box that bridges the gap.

Each port is a shape on its own layer: port 1 on layer 201, port 2 on layer 202, and so on. Its settings (direction, layers, impedance) are stored on the shape, so they are saved with your layout.

6.2 Placing a port with the EM Port tool

1. Press **EM Port** in the toolbar.
2. Move the mouse over the edge of the metal where the port should be. The edge under the cursor is highlighted.
3. Click. The port shape is placed, and the port dialog opens.
4. Check the settings and press **OK** (**Cancel** removes the port again).

For an in-plane port, hold **Shift** and click into the gap between two shapes of one metal.

The tool picks only metals you can see. Hide a layer in the layer panel, and its edges are ignored, so you can reach a metal edge below a larger shape. If the only edge under the cursor is on a hidden layer, the status bar tells you. Hiding a layer does not limit the port itself: the port dialog still offers every metal drawn there, e.g. the ground plane you hid to see the pad.

In the cascode, the two RF pads get a via port each, at the outer edge of the pad (port 1 at the Cas_out pad):